Classification of Robotic Sensors

Proprioceptive vs. Exteroceptive Sensing Systems

Overview of Sensor Types



Internal Sensors

Proprioceptive sensors that monitor the robot's internal state, such as joint angles, velocity, and battery levels.



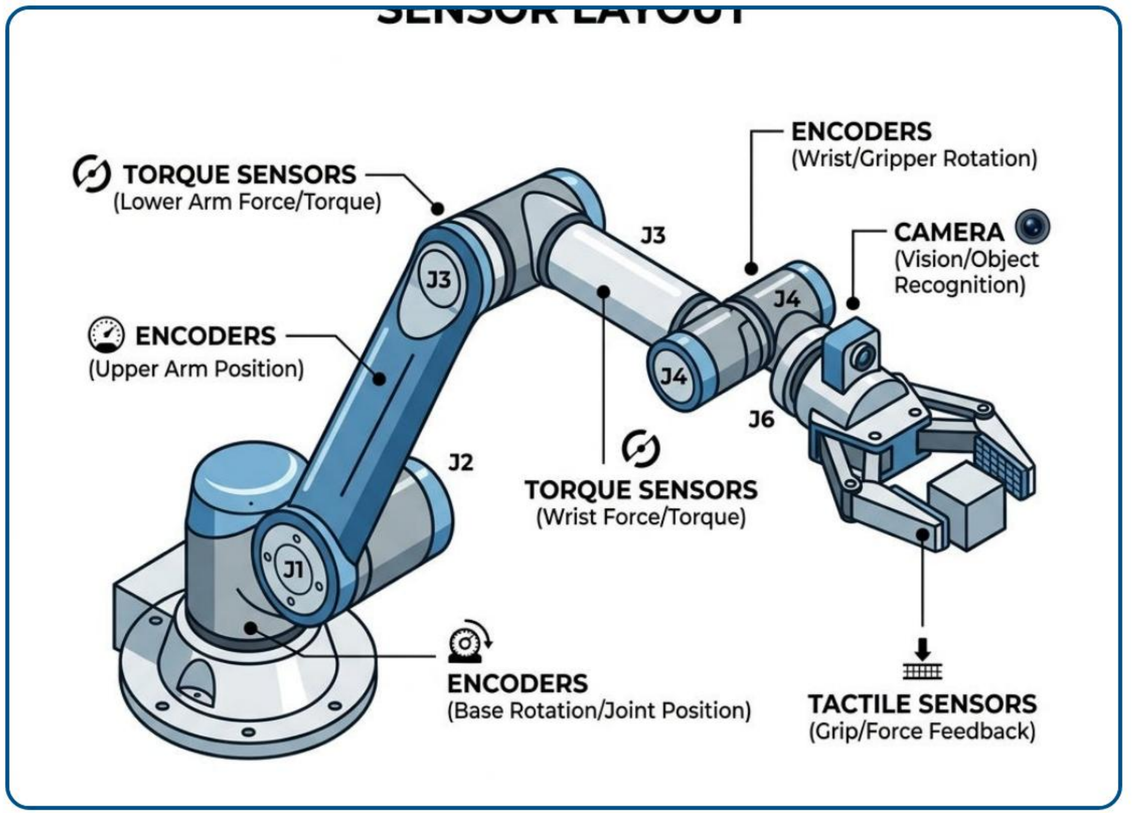
External Sensors

Exteroceptive sensors that perceive the environment, categorized into contact and non-contact types.

Internal (Proprioceptive) Sensors

Primary Functions

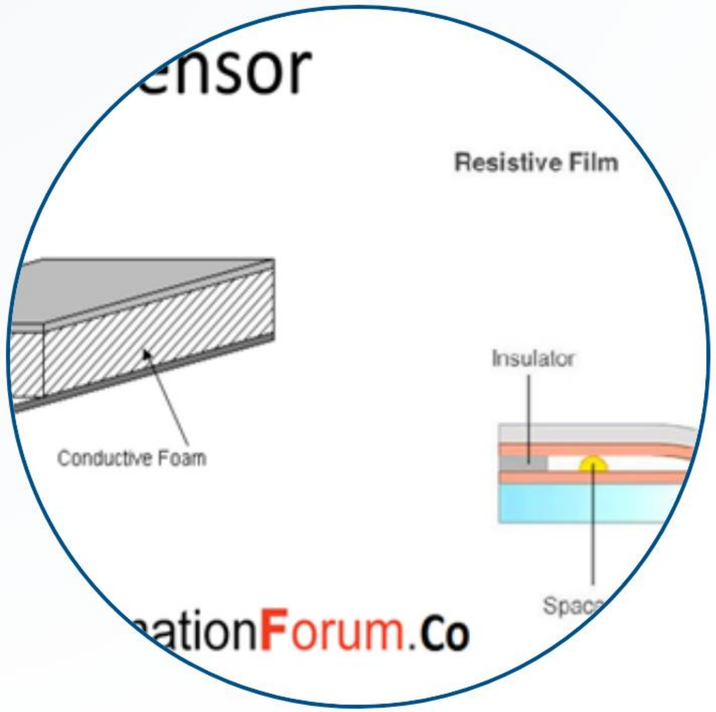
- **Location Sensors:** Potentiometers and Encoders for joint position.
- **Velocity Sensors:** Tachometers measuring rotational speed.
- **Acceleration Sensors:** Accelerometers measuring dynamic changes.
- **Force Sensors:** Strain gauges and Piezoelectric load cells.



External Sensors

Exteroceptive Perception of Surroundings

External: Contact Type Sensors



Sensing via Interaction

These sensors require physical interaction with the object or medium to gather data.

Thermocouple: Measures temperature through junction voltage.

Pressure Sensors: Monitors applied force per unit area.

Touch Sensors: Tactile feedback for object manipulation.

External: Non-Contact Type

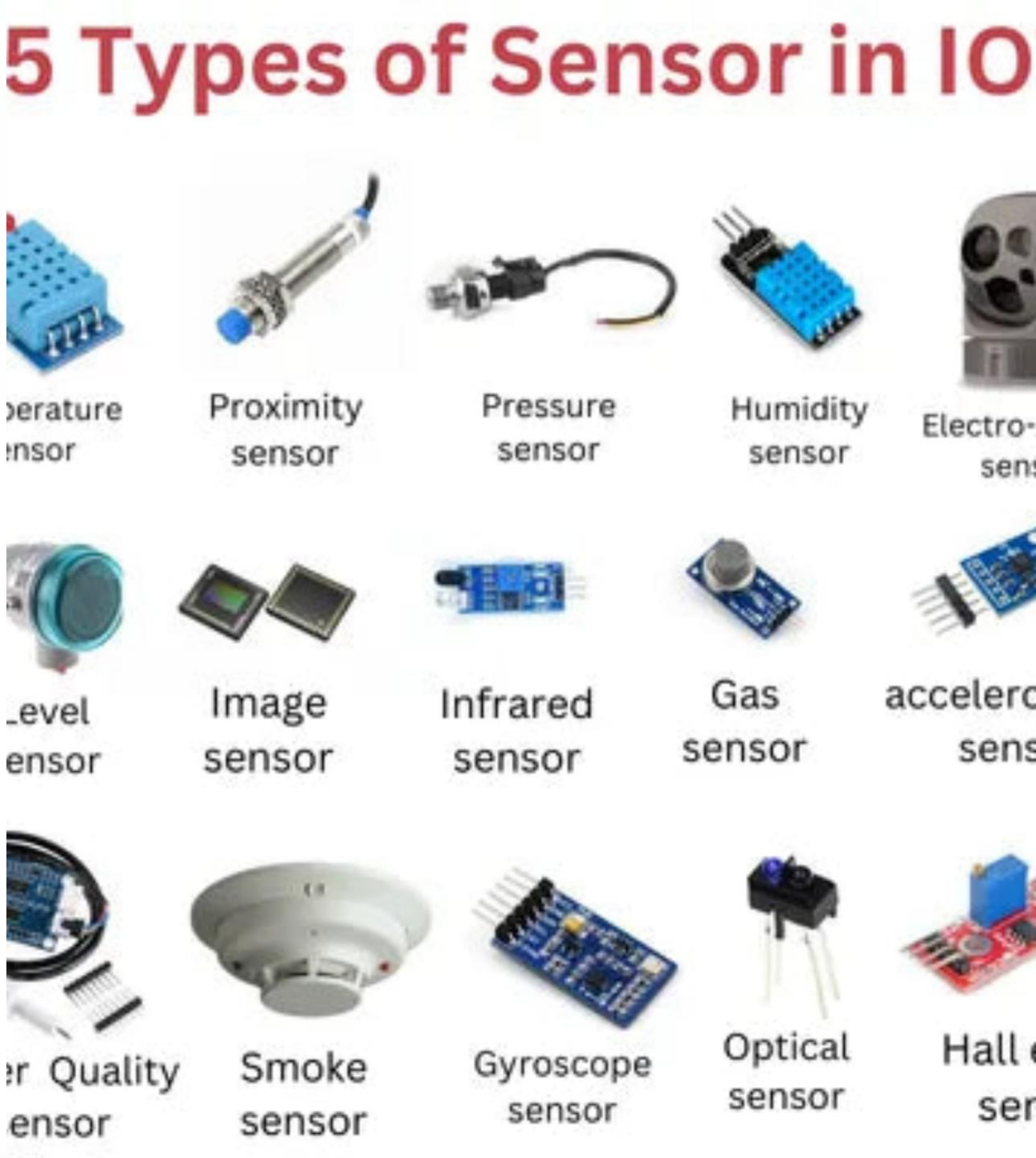
Remote Perception

Non-contact sensors rely on electromagnetic or acoustic waves to detect environmental properties from a distance.

Infrared (IR): Proximity and range sensing via light reflection.

Gas Sensors: Detection of molecules like Carbon Monoxide.

Light Sensors: Photodiodes measuring ambient intensity.



Summary of Classifications

Category	Sub-Type	Key Examples
Internal	Proprioceptive	Location (Encoders), Velocity (Tachometers)
	Kinematic	Acceleration (IMU), Force (Load Cells)
External	Contact	Thermocouple, Pressure, Tactile/Touch
	Non-Contact	Infrared, Gas (CO), Light (Photodiode)

Questions?

Thank you for your attention.

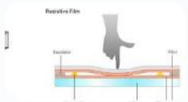
[End of Sensor Classification Deck]

Image Sources



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